Angel Patricio

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Education

Massachusetts Institute of Technology — **Cambridge, MA**
M.Eng., Electrical Engineering and Computer Science — *Expected 2027–2028*

Massachusetts Institute of Technology — **Cambridge, MA**
B.S., Artificial Intelligence and Decision Making — *2022–2026*

Coursework (*grad): Deep Learning*, Reinforcement Learning*, Statistical Learning Theory*, Nonlinear Optimization*, NLP, Real Analysis

Publications

J. Yuan, B. Le, K. Le, Y. Shi, A. Singh, R. Sharma, **A. Patricio**, R. Raskar. SONAR Web: A Platform-Agnostic Framework for Real-Time Decentralized Learning Across Heterogeneous Edge Clients. *ACM FLEdge-AI Workshop*, 2025. **Best Presentation Award.**

Experience

Selected Projects

A Recipe for Grokking with Recurrent-Depth Models — **aopatric/recurrent-grokking**
Independent Research — *Oct 2026–*

Ablating the necessary training setup for looped transformers to grok multi-hop addition mod p. Ongoing with an existing WIP writeup. [blog]

- Designed and trained fixed-depth, universal-transformer and Huginn-style toy-scale models on a synthetic data. Ablated Huginn and the UT toward each other, measuring generalization performance and stability over both input values and chain length.
- Built with PyTorch; analyzed artifacts with Fourier fits and recorded Neel Nanda's progress measures for grokking on a 8x 5090 node.

Intact: Interpretable Activations for Reward Hacking — **aopatric/intact**
Independent Project — *Aug 2026–*

Built a library to extract activations at and around the site of hacking behavior in a canonical Qwen3-4B setup from a MATS alum.

- Designed a CLI and Python API to sample rollouts, grade them with upstream's built-in grader, and extract the activations from specified hook points near the token where hacking begins. Reproduced upstream's hack rate with the new tooling (73.7% vs. 79%).
- Sampled with vLLM; measured PyTorch extraction to within 4e-5 relative vs. generation-time values. Stored activations as fp16 Parquet.

The Edge of Stability for LLM Post-Training — **aopatric/post-training-EoS**
Course Project, MIT 6.7960 Deep Learning — *Fall 2025, Revisited 2026*

Explored whether the Edge of Stability appears when post-training LMs, then built new tooling after auditing my original null result. [blog]

- Traced my original null result to a noisy curvature proxy and a clipping error, then rebuilt the pipeline. Saw signs of EoS on a 'post-trained' MLP but no clear confirmation on Pythia-70m, so reported the results as inconclusive in a blog post.
- Built GPU-native Hessian tooling (294 tests, checked on exact Hessians); found fp32 updates lost a median 3.4% of each step to rounding.

Academic Research

A Benchmark for Spatial Reasoning with Origami — **Cambridge, MA**
MIT Computer Science and AI Lab (CSAIL), Solar-Lezama Group — *Feb 2025–Aug 2025*

Helped design an inverse-origami benchmark for spatial reasoning in ML models; acknowledged in OrigamiBench (ICML LM4Plan '26).

- Conducted an extensive literature review on spatial-reasoning evaluation for neuro-symbolic and language models; presented findings and benchmark ideas at weekly planning meetings.
- Adapted a published origami environment with an .obj-to-.FOLD target pipeline (25 targets); ran search and RL baselines.

SONAR Web: Real-Time Decentralized Learning Across Heterogeneous Edge Clients — **Cambridge, MA**
MIT Media Lab, Camera Culture Group — *Nov 2024–Feb 2025*

Built the first clients for privacy-preserving decentralized training across devices, later extended by the team and published.

- Wrote peer-to-peer logic for Python and JavaScript clients over the group's WebRTC layer, streamed weights in chunks past the 16MB limit, and proposed a JSON weight format; validated up to 12 peers on 4 topologies.
- Implemented in PyTorch, TensorFlow.js and WebRTC; trained MLPs and CNNs on MNIST and CIFAR across platforms.

Masked Language Modeling for Genomics — **Cambridge, MA**
MIT Institute for Medical Engineering and Science (IMES), Edelman Lab — *Apr 2024–Aug 2024*

Scaled the lab's DistilBERT masked-LM pipeline for single-cell expression from Tabula Sapiens to 30M+-sample CELLxGENE.

- Adapted preprocessing to the census (missing values, new featurization), moved training to multi-GPU cloud instances, cutting epoch time from over an hour to about 15 minutes, and trained several model variants.
- Used HF Transformers and Accelerate (PyTorch DDP) on Vast.ai; built tooling to launch and log WandB sweeps in one call.

Technical Skills

ML: MLPs, CNNs, Transformers, LLMs, Deep RL, LoRA
Frameworks: PyTorch, JAX, scikit-learn
Languages: Python, JavaScript, Julia
Interpretability: Probing, Patching, Steering
Libraries: NumPy, transformers, verl, TransformerLens
Tools: Git, Bash, Vast.ai, vLLM, TensorBoard, uv